ML Challenge 2025: Smart Product Pricing Solution

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1. Executive Summary

The solution employs a **multimodal deep learning fusion strategy** using a **LightGBM** regressor. We extract high-quality features from product text (MiniLM embeddings), images(EfficientNet features), and data (TF-IDF, Target Encoding) before combining them via for the final model. This approach effectively captures product value drivers across all data types, yielding a robust prediction of the target.

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2. Methodology Overview

2.1 Problem Analysis

The challenge is to accurately predict product price given catalog content (text), image link, and a small set of information, optimizing for the **SMAPE** metric, which heavily penalizes large relative prediction errors.

Key Observations:

- **Target Distribution:** The target is heavily right-skewed, necessitating a logarithmic transformation (ln(price + 1)) and **winsorization** to stabilize the model.
- **Multimodal Nature:** Text and Images are the primary sources of intrinsic product value and detail; ignoring either leads to significant information loss.

- **Categorical Leakage:** is a key feature, but must be handled carefully with **Out-of-Fold Target Encoding** (GroupKFold on Brand) to prevent data leakage and overfitting.

2.2 Solution Strategy

The high-level strategy is **Feature Engineering -> Multimodal Feature -> Extraction -> Feature Fusion -> Group-Agnostic Regressor Training.**

Approach Type: Hybrid (Deep Learning Feature Extraction + LightGBM Regressor)

Core Innovation: Seamless, Cache-Only Fusion of State-of-the-Art EfficientNest (Vision) and MiniLM (Text) Embeddings with Traditional TF-IDF and Target-Encoded Features, trained using GroupKFold cross-validation.

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3. Model Architecture

3.1 Architecture Overview

The architecture is a **two-stage** process:

1. **Feature Generation Stage:** Parallel extraction of 4 feature sets (Vision, Text, Tabular/Engineered, TF-IDF).
2. **Prediction Stage: LightGBM** Regressor trained on the concatenated (fused) feature vector.

$$\mathbf{X}_{\text{final}} = \text{hstack}([\mathbf{X}_{\text{Vision}}(\text{EfficientNet}),\ \mathbf{X}_{\text{Text}}(\text{MiniLM}),\ \mathbf{X}_{\text{Engineered}}(\text{IPQ, Brand_TE}),$$

$$\mathbf{X}_{\text{Sparse}}(\text{TF-IDF})]) \quad \rightarrow \quad \text{LightGBM} \quad \rightarrow \quad \hat{y}_{\log}$$

3.2 Model Components

Text Processing Pipeline:

- **Preprocessing steps: Tokenization and Truncation** (Input text is tokenized into subwords and truncated to max_length = 128 tokens for model input.)
- **Model type: Sentence-Transformers/all-MiniLM-L6-v2** (Transformer-based Sentence Embedding Model).
- **Key parameters:** Max-length = 128; Output dimension: 384; Mean Pooling for sentence representation.

Image Processing Pipeline:

- **Preprocessing steps:** Resize (224, 224); ToTensor; ImageNet Normalization.
- **Model type: EfficientNet-B0** (timm implementation).
- **Key parameters:** Pretrained on ImageNet; num_classes = 0 (feature extractor mode); Output dimension: 512.

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4. Model Performance

4.1 Validation Results

- **SMAPE Score: 53.555 (previous validated SMAPE score)**
- Other Validation Metrics:
  1. MAE (Mean Absolute Error): 12.3133
  2. RMSE (Root Mean Squared Error): 22.0695
  3. R^2 (Coefficient of Determination): 0.2588

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5. Conclusion

This solution successfully implements a powerful Hybrid architecture, leveraging Transfer Learning from pre-trained

Vision and Text models to extract high-dimensional features, which are then optimally weighted by a LightGBM model. The use of GroupKFold and Target Encoding mitigates crucial data leakage issues. The entire pipeline is engineered for efficiency, relying on a **local image cache** for fast feature generation, demonstrating a robust and high-performing smart product pricing system.

Appendix

A. Code artefacts

Google Drive Link:
https://drive.google.com/file/d/1X5tlYcFztTGkHrQI9yMkIa6ubr6hFsiF/view?usp=sharing

B. Additional Results

- **Average Best Iterations:**

  1000. The final model was trained for boosting rounds, utilizing the full allocated budget as early stopping was not triggered in any fold.

- **Feature Contributions:**

  The fusion approach implies all 4 feature sets (Vision, Text , Tabular, TF-IDF) contribute to the final predict.